

9

COMPREHENSION OF DISCOURSE

Once a reader has parsed and interpreted a sentence, his or her work is probably not done. The reader must understand the text that includes the sentence; the meaning of a coherent text is more than the sum of the individual sentence meanings. The reader must knit the sentences together so that the propositions they contain can be combined as the writer intended. The reader must make inferences to fill in material that is only implicit in the text itself, and in doing this the reader often has to appeal to knowledge about the world, including the rest of the discourse. Ideally, the reader will understand the text in the sense of knowing what kind of real-world situation it is describing, or knowing how to do what the text is instructing the reader to do, or believing (or disbelieving) the claims it contains, and so on. Sometimes, the reader's prior knowledge distorts the text's message (Bartlett, 1932); nonetheless, what stays with the reader is generally the "gist" of the text.

Language contains devices that facilitate (but do not guarantee) accurate understanding of the text's message. Some of these devices become apparent when a text fails to use them. Consider the following malodorous passage:

She set it down. Sue had gone to Kroger's to buy a bird. The day before, a mother had asked her to buy the turkey. The table's leg's collapsed. It had to be big enough to feed all of them. But it was too big for a table. She cleaned them from the floor. (9.1)

There are so many things wrong with this passage! The first occurrence of *she* has no apparent referent; the second sentence talks, without warning, about an event that took place earlier than the preceding sentence; the third sentence talks oddly about *a*, not *her*, mother, and is the mother talking specifically about the turkey that Sue bought at Krogers? That's odd. And what made the table's leg collapse? And on and on.

We'll soon be discussing how people interpret pronouns and other anaphors, how they make inferences that go beyond the text, how they understand the sequence of events in an episode, and in general, how they glue a text together. But first, consider the following passage from Bransford and Johnson (1972):

The procedure is actually quite simple. First you arrange things into different groups. Of course, one pile may be sufficient depending on how much there

is to do. If you have to go somewhere else due to lack of facilities, that is the next step, other wise you are pretty well set. It is important not to overdo things. That is, it is better to do too few things rather than too many. In the short run this may not seem important but complications can easily arise. A mistake can be expensive as well. At first the whole procedure will seem complicated. Soon, however, it will become just another facet of life. It is difficult to foresee any end to the necessity for this task in the immediate future, but then one can never tell. After the procedure is completed one arranges the materials in groups again. Then they can be put into their appropriate places. Eventually they will be used once more and the whole cycle will have to be repeated. However, this is part of life. (9.2)

While you can understand the meaning of each sentence (in some sense) and even put the sentences together, the passage as a whole has no clear meaning. But if you were first told that it is describing doing your laundry, it becomes meaningful. This indicates that comprehending a text usually requires the reader to use real-world knowledge that includes plans, intentions, and causation. We will shortly discuss how readers integrate the text that they read with their knowledge of the real world and use both together to create a situation model or mental model that the reader takes as the meaning of the text. But first we will rather briefly discuss some analyses of what kind of representation readers have of a text that they read successfully.

Discourse representations

Kintsch's representations of discourse

Much of the experimental research on text comprehension was stimulated by a theory that Kintsch and his colleagues developed over a long period of time (Kintsch, 1974, 1988, 1994, 1998; Kintsch & van Dijk, 1978; van Dijk & Kintsch, 1983). One of the primary contributions of this theory was to distinguish between different mental representations of a text. These representations are called the surface representation, the propositional representation, and the situation model (Kintsch, 1994). The previous chapter focused on the first two representations, discussing how a reader goes from the printed text to a parsed representation of the form of a sentence (the surface representation) and then to a meaningful interpretation of the sentence (the propositional representation). Text and discourse processing researchers concentrate on how a reader puts the propositional representations of sentences in a text together to arrive at a representation of the situation that the text is describing (the situation model). In some versions of this theory, a distinction is made between “microstructure” and “macrostructure.” Microstructure is a level of representation that puts the propositions together into a connected structure; macrostructure is the gist of the text, an edited version of the microstructure tied into schemas from long-term memory.

The typical way of representing a proposition, in approaches like Kintsch's, is as a predicate followed by its arguments. We will illustrate how this is done by sketching part of an analysis of the paragraph shown in Table 9.1. A predicate can be represented by the meaning of a verb (where meaningful concepts rather than words are indicated by UPPERCASE) and the arguments can be represented by the meanings of the nouns or prepositional phrases or adverbs that participate in the event denoted by the verb. For example (TEACH, SPEAKER, STUDENT) represents the proposition that some speaker is teaching some student(s). Other propositions represent adjective–noun combinations such as (VIOLENT, ENCOUNTER) and quantifier–noun combinations such as (ALL, STUDENT). Others represent ideas in phrases such as (OF REALITY, VOICE), others conjoin concepts such as (AND, STUDENT, SPEAKER), and others express location and time

TABLE 9.1 A paragraph and the propositions extracted from it

A series of violent, bloody encounters between police and Black Panther Party members punctuated the early summer days of 1969. Soon after, a group of Black students I teach at California State College, Los Angeles, who were members of the Panther Party, began to complain of continuous harassment by law enforcement officers. Among their many grievances, they complained about receiving so many traffic citations that some were in danger of losing their driving privileges. During one lengthy discussion, we realized that all of them drove automobiles with Panther Party signs glued to their bumpers. This is a report of a study that I undertook to assess the seriousness of their charges and to determine whether we were hearing the voice of paranoia or reality.

<i>Proposition number</i>	<i>Proposition</i>
1	(SERIES, ENCOUNTER)
2	(VIOLENT, ENCOUNTER)
3	(BLOODY, ENCOUNTER)
4	(BETWEEN, ENCOUNTER, POLICE, BLACK PANTHER)
5	(TIME: IN, ENCOUNTER, SUMMER)
6	(EARLY, SUMMER)
7	(TIME: IN, SUMMER, 1969)
8	(SOON, 9)
9	(AFTER, 4, 16)
10	(GROUP, STUDENT)
11	(BLACK, STUDENT)
12	(TEACH, SPEAKER, STUDENT)
13	(LOCATION: AT, 12, CAL STATE COLLEGE)
14	(LOCATION: AT, CAL STATE COLLEGE, LOS ANGELES)
15	(IS A, STUDENT, BLACK PANTHER)
16	(BEGIN, 17)
17	(COMPLAIN, STUDENT, 19)
18	(CONTINUOUS, 19)
19	(HARASS, POLICE, STUDENT)
20	(AMONG, COMPLAINT)
21	(MANY, COMPLAINT)
22	(COMPLAIN, STUDENT, 23)
23	(RECEIVE, STUDENT, TICKET)
24	(MANY, TICKET)
25	(CAUSE, 23, 27)
26	(SOME, STUDENT)
27	(IN DANGER OF, 26, 28)
28	(LOSE, 26, LICENSE)
29	(DURING, DISCUSSION, 32)
30	(LENGTHY, DISCUSSION)
31	(AND, STUDENT, SPEAKER)
32	(REALIZE, 31, 34)
33	(ALL, STUDENT)
34	(DRIVE, 33, AUTO)
35	(HAVE, AUTO, SIGN)
36	(BLACK PANTHER, SIGN)
37	(GLUED, SIGN, BUMPER)
38	(REPORT, SPEAKER, STUDY)

(Continued overleaf)

TABLE 9.1 (Continued)

<i>Proposition number</i>	<i>Proposition</i>
39	(DO, SPEAKER, STUDY)
40	(PURPOSE, STUDY, 41)
41	(ASSESS, STUDY, 42, 43)
42	(TRUE, 17)
43	(HEAR, 31, 44)
44	(OR, 45, 46)
45	(OF REALITY, VOICE)
46	(OF PARANOIA, VOICE)

Lines indicate sentence boundaries. Propositions are numbered for ease of reference and a number in a propositional argument refers to the proposition with that number. (The same numbers appear in Figure 9.1.) Adapted from Kintsch and Van Dijk (1978).

(TIME: IN, ENCOUNTER, SUMMER) or (LOCATION: AT, CAL STATE LOS ANGELES). In all cases the notation not only conveys that the concepts go together but contains some of the meaning. In (TEACH, SPEAKER, STUDENT), for example, TEACH is understood to be the verb, the second argument, SPEAKER, the agent, etc. See Table 9.1 for more examples of propositions.

As you can see in Table 9.1, a single sentence is usually broken down into several propositions. However, propositions can be built recursively so that a proposition can have other propositions as arguments. For example, in the proposition (COMPLAIN, STUDENT, 19), 19 stands for the proposition (HARASS, POLICE, STUDENT). Thus, by this recursive definition, a proposition can stand for a fairly complex thought. As you can also see in Table 9.1, many propositions are of this recursive type.

The microstructure of the text in Table 9.1 is a structure built out of the list of propositions in that table, and can be represented as a graph as in Figure 9.1. The graph is intended to capture the coherent relations among the propositions. Its structure is, in Kintsch’s work, largely determined by how the propositions share concepts. Propositions with the same concept are said to overlap, and if the proposition that is being processed at one point in time shares a concept from a different proposition that is still in short-term memory, those two propositions can be linked together in the microstructure graph, from Kintsch and van Dijk (1978).

We will not review in detail the process by which Kintsch and van Dijk claim that such a coherence graph is created nor how a macrostructure is built up by attaching “important” aspects of the microstructure to a pre-existing “schema” (e.g., one’s knowledge of what is likely to happen in a fairytale or a narrative about a visit to a restaurant). Suffice it to say that early research provided support for some of the predictions of the model, using measures of memory, of reading time, and of reaction time to probes, among other experimental methods. One prediction from their model is that propositions that overlap with many other propositions will appear high in the microstructure graph, and thus they are remembered better than lower-level propositions (e.g., Kintsch, Kozminsky, Streby, McKoon & Keenan, 1975; Meyer, 1975). A second prediction is that the reading time for a passage with more propositions is greater than for a passage with fewer propositions even when the number of words in the two passages is equal (Kintsch & Keenan, 1973).

A particularly interesting test of the Kintsch and van Dijk model was conducted by McKoon and Ratcliff (1980). Since this study is interesting methodologically as well, we will describe it in some detail. They used a variation of the priming technique (described below) to get at the structure of discourse memory. In one experiment, participants saw “paragraphs” like the following:

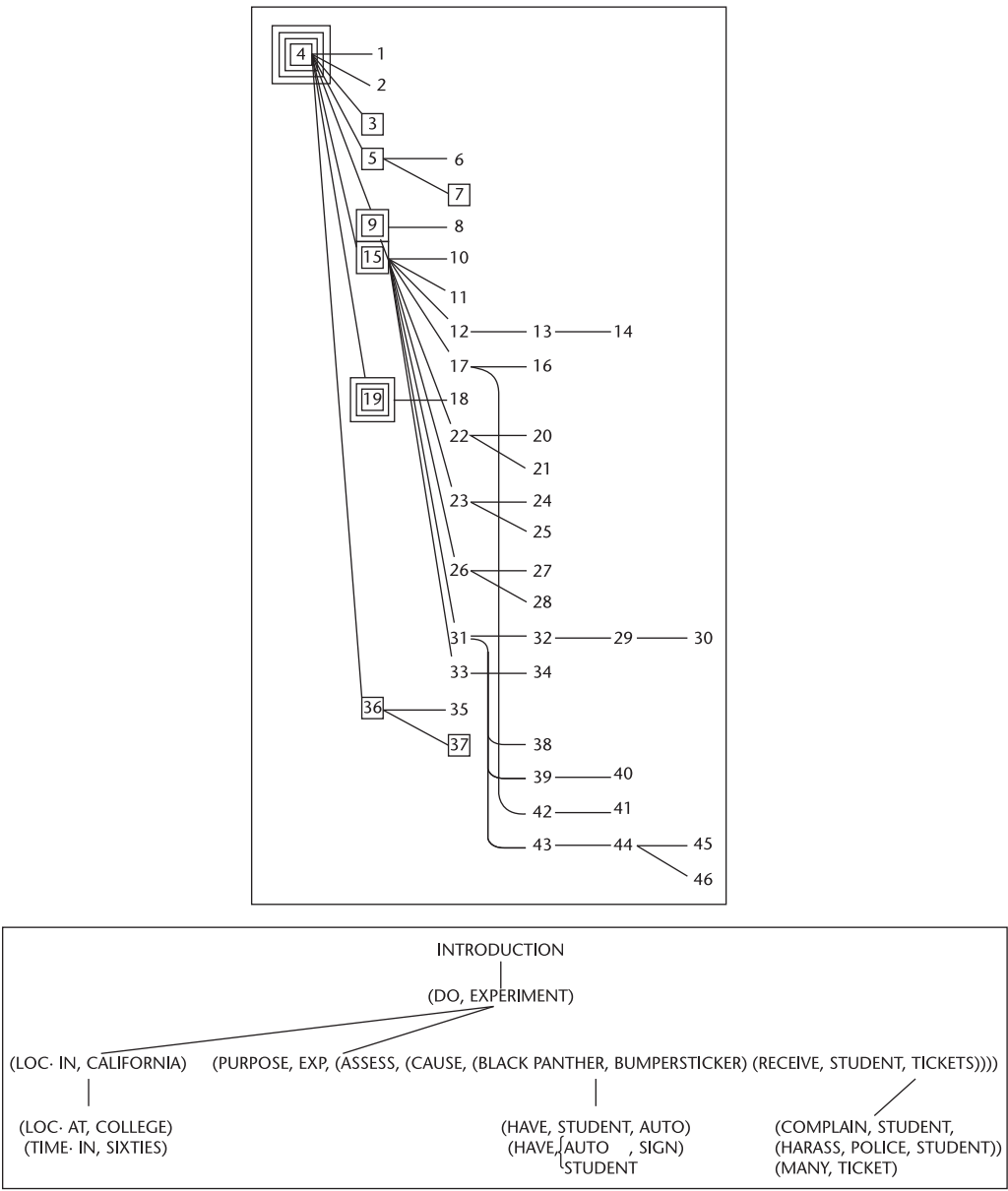


FIGURE 9.1 (Top) The complete “coherence graph” representing the microstructure of the “Bumperstickers” paragraph of Table 9.1. The numbers represent propositions and the number of squares around a proposition indicate the number of cycles it was held in STM for processing. (Bottom) The macrostructure for the Bumperstickers passage. The basic schema is a “scientific report” schema and the important propositions from the microstructure are incorporated into it. Reproduced from Kintsch and Van Dijk (1978) with permission from the American Psychological Association and the authors

- (1) The businessman gestured to a waiter.
- (2) The waiter brought coffee.
- (3) The coffee stained the napkins.
- (4) The napkins protected the tablecloth.

- (5) The businessman flourished documents.
 (6) The documents explained a contract.
 (7) The contract satisfied the client. (9.3)

A propositional analysis of this paragraph is presented in Figure 9.2. The main feature of Figure 9.2 to notice is that Proposition 2 is closer to Proposition 5 in the microstructure than Proposition 4 is to Proposition 7 even though they are equally far apart in the surface representation of the text. Thus, the concepts WAITER and DOCUMENTS are closer in the microstructure than are the concepts NAPKINS and CLIENT (even though the two pairs of words are equally far in the surface text and the individual words in sentence 2 are not related to those in sentence 5). McKoon and Ratcliff tested for closeness using a recognition memory procedure. After reading the paragraph, participants saw a series of probe words, which they were to judge as having been present in the paragraph or not. (Their responses were timed.) Of primary interest was the effect of one probe word (the prime) on the response time to the following probe word (the target). The logic is that if responding to one probe word speeds up responding to a second probe word, then the words are closely linked in some memory structure, just as the phenomenon of *cat* priming *dog* in a lexical decision task argues that CAT and DOG are closely linked in some sort of memory structure. McKoon and Ratcliff found that *waiter* primed *documents* more than *napkins* primed *client* indicating that WAITER and DOCUMENTS were more closely linked in memory than NAPKINS and CLIENT (as predicted by the Kintsch and van Dijk model).

A problem with the above experiment is that the text was not very natural. Accordingly, McKoon and Ratcliff used more natural paragraphs such as the following:

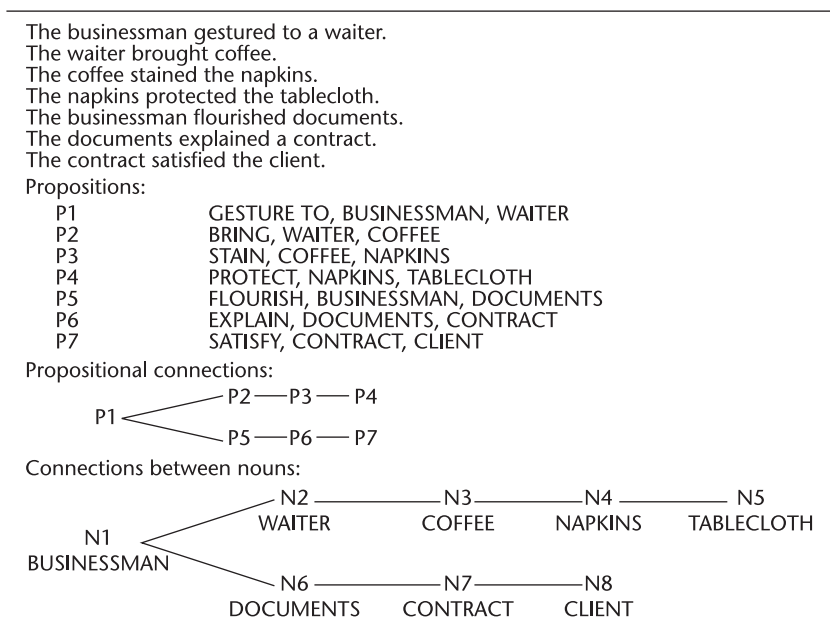


FIGURE 9.2 A paragraph from McKoon and Ratcliff (1980) together with the propositional analysis and propositional structure. Reproduced with permission from Academic Press and the authors

Early French settlements in North America were strung so thinly along the major waterways that land ownership was a problem. The Frenchmen were fur traders, and, by necessity, the fur traders were nomads. Towns were few, forts and trading posts were many. Little wonder that the successful fur trader learned to live, act, and think like an Indian. Circulation among the Indians was vital to the economic survival of the traders. (9.4)

The probe task in the second experiment was also more meaningful, in that participants were given sentences such as *The fur traders were nomads* and asked to respond whether the idea was stated in the paragraph. (The probes essentially tested for memory of propositions.) Again, McKoon and Ratcliff looked at the effect of one probe on the next, and contrasted the priming effects of propositions that were near in the surface but far in the propositional structure with those that were near in the propositional structure but far in the surface. They found, for example, that *Circulation among the Indians was vital* which was near *The fur traders were nomads* in the propositional structure but far in the surface primed it more than *Land ownership was not a problem* which was near *The fur traders were nomads* in the surface but far in the propositional structure.

While all of these demonstrations are of interest in establishing the model, none provides definitive evidence in favor of it. The model does not have a fully adequate algorithm for creating a microstructure. Creation of a microstructure depends to a great extent on an experimenter's intuition. It is not enough to use algorithms that place propositions high in a microstructure simply because they share a noun phrase. Such algorithms may go wrong if the same noun phrase refers to different entities; propositions that share a noun phrase may involve different referents, and should not play a central role in the microstructure. Thus a successful demonstration that memory is better for propositions high in a microstructure may largely be due to the experimenter's intuitions about the importance of various propositions rather than to any intrinsic feature of the model. Similarly, the fact that reading time increases with the number of propositions could be due to the greater syntactic complexity and denseness needed to express more propositions with the same number of words. In summary, Kintsch and van Dijk's theorizing opened up the field of text processing to objective experimental study. Their representations of text captured some important aspects of a text (e.g., how propositions are linked together and how some are more central than others). Experimental work has provided support for many of the claims of the theory (e.g., the claim that some propositions are more central than others, that some propositions are closely linked to other, and that some sentences are easier to integrate into the structure of a text than others). Many of the core processing claims of the model are still widely accepted: readers are generally thought to encode the basic ideas (the propositions) in a text and to connect them together into more global, integrative structures using a limited-capacity processing system that forces integration to be selective. More recent work (some of which is discussed in the following pages) explores the details of how memory processes are involved in comprehending discourse and how the reader uses a rich knowledge base to guide comprehension (Gerrig & McKoon, 1998; Myers & O'Brien, 1998; Sanford & Garrod, 1998, 2005).

Schemas, scripts, and story grammars

It is interesting to compare the theories of sentence structure, discussed in the previous chapter, with the theories of discourse structure discussed in this chapter. The sentences of a language are described, in great detail, by the grammar of that language. Some sequences of words are grammatical; some are simply not sentences at all. Linguists have provided detailed rules that generate the sentences of a language—contentious rules, to be sure, but detailed and precise nonetheless. Psycholinguists have

developed and tested explicit theories of how these rules are used and how they are integrated with nonlinguistic knowledge—contentious theories, again, but detailed and explicit.

Early discourse researchers attempted to analyze discourses in the way that linguists have analyzed sentences. They attempted, for instance, to devise grammars that would specify the permissible and impermissible structures of stories (e.g., Rumelhart, 1975; Thorndyke, 1977). Some rules specified the possible sequence of subsections of a story: SETTING, THEME, PLOT, RESOLUTION. Other rules specified how the plot subsection was to be divided into subsections, with such plot devices as goals and resolutions. Experimental research showed that disrupting the preferred order of a story (e.g., putting the theme last instead of first) disrupted memory for the story, and showed that if a superordinate proposition in a story were forgotten, its subordinates would tend to be forgotten (Thorndyke, 1977). However, most researchers now agree that while stories do have structure and violating the structure makes them less easy to comprehend, this structure is not like the grammar of a language. There is no clear line between a story and a non-story, as there is between a sentence and a non-sentence. Further, there is no evidence that readers of a story give priority to any story grammar rules in the same way that they seem to give priority to the rules of a sentence grammar; while they cannot legitimately understand *man bites dog* to mean that a dog bit a man, they can puzzle out stories presented in fascinatingly twisted ways.

A related approach to thinking about the structure of discourses and how people can use that structure to understand a text falls under the heading of schema theory (see the earlier discussion of Bransford & Johnson's 1972 "washing clothes" demonstration for an example). The fundamental claim of schema theories is that we have pre-stored, possibly elaborate, knowledge structures for different situations that guide how all the information in the text is understood and stored in memory. For example, in the laundry passage at the beginning of the chapter, the assumption of schema theorists is that you understand the passage (given the title *Laundry*) because the title allows you to retrieve a schema from your real-world knowledge for doing laundry. Once this schema is retrieved, you then tie the information in the passage into this schema.

A schema not only gives us the possibility of tying the propositions of a text down to a concrete type of situation; a schema can also let us fill in information that the writer of a text does not bother to include. Consider one specific example of a type of schema, namely the "script." A script is one's representation of a more-or-less stereotyped, socially sanctioned, series of actions appropriate to a given setting. Consider the restaurant schema or restaurant script (Bower, Black, & Turner, 1979; Schank & Abelson, 1977). When you recognize that a discourse is about visiting a restaurant, a restaurant script, essentially a structured sequence of events in a meal at a restaurant, is retrieved from memory, and certain "default values" for what happens are retrieved. The default values are presumably the actions that occur in your normal restaurant experience. The information from the script is then incorporated into your understanding of the story so that if the first sentence is about entering the restaurant and the second sentence is about ordering food, the script would fill in the missing steps, such as waiting to be seated, sitting down, getting menus, etc. Information that is redundant with the script (e.g., that the restaurant had menus) might not be explicitly added to the representation, but information that is inconsistent with the script (e.g., that the restaurant did not have menus) might be highlighted as being informative.

One of the reasons that schema theories attracted interest was that they appeared at a time when psycholinguistics had pretty much ignored the role of real-world knowledge in understanding discourse. A major thrust of these theories was to demonstrate that comprehension of a text does not follow automatically from knowing the literal meaning of its sentences, but instead has to rely on real-world knowledge such as what it is actually like to do laundry, why men go to jewelers, etc. But nobody has an adequate explicit theory of real-world knowledge, and it is not likely that one

will appear in the foreseeable future. No “grammar of knowledge” is available. In retrospect, while schema theories probably captured something important and real about how we understand discourses, the predictions they made basically boil down to claiming that we are best at understanding what makes sense to us, and that what makes sense depends on what we know. (For an excellent discussion and critique of schema theories, see Alba & Hasher, 1983.)

Putting sentences together

Let us turn to a less ambitious project than explaining what knowledge is. Psycholinguists can say quite a lot about what linguistic devices make a text coherent and how readers use these devices. These devices let the reader know when something already familiar in the text is being referred to, and when something new is being introduced. They let the reader know the sequence of events and the causal relations among them. They support economy and efficiency—don’t waste time saying what doesn’t need saying—and, used properly, they can help the reader build a mental model of the situation being written about.

Pronominal anaphora

A third-person pronoun used in a text usually refers to an entity that has already been introduced in the text (which is termed anaphoric reference). Such pronouns contribute to the coherence of a text; they help tie the sentences in a text together. A reader’s task upon encountering a pronoun is to determine what it refers to (see Garnham, 1999, for a thorough discussion).

When a pronoun has an antecedent in the text, the position and role of the antecedent can affect how a reader determines the referent of a pronoun. Early work showed that pronouns are read more easily when their antecedent is closer than when it is more distant in the text. Clark and Sengul (1979) found that as the distance between the pronoun and its antecedent increased, the reading time of the sentence containing the pronoun increased. Ehrlich and Rayner (1983) also varied the distance between pronoun and antecedent, and attempted to measure the time course of the effect by examining the pattern of eye movements on the sentence that contained the pronoun. There were three levels of distance: near, when the antecedent noun was the word immediately before the pronoun; intermediate, where about a line of text intervened; and far, where at least three lines of text intervened. A central purpose of the experiment was to test the immediacy hypothesis of Just and Carpenter (1980, see Chapter 4), which states that all linguistic operations, such as finding the antecedent of a pronoun, are completed before the eyes move on to the next word. A major problem in this research is identifying the fixation on which the pronoun is encoded, since it is usually fixated much less than half the time. Since the perceptual span extends to the right of the fixated word but not to the left of it (see Chapter 4), Ehrlich and Rayner assumed that if the pronoun was not fixated, then it was encoded on the fixation to the left of the pronoun. That is, the “encoding” fixation for a pronoun was either on the pronoun itself, or if it was not fixated on, the prior word.

There were two major findings. First, the fixation time on the encoding fixation was significantly longer (about 20 ms) than the previous one (in all distance conditions), suggesting that the reader was immediately starting to search for the antecedent in all conditions. However, in the immediate and near conditions, the two fixations following the encoding fixation returned to normal length, whereas these next two fixations in the far condition got even longer. The former finding is only suggestive, because the words being fixated are different. However, the latter finding strongly suggests that the process of searching for the antecedent may begin immediately, but in the case of the far condition, the search often continues for another fixation or two. This rules against

a strong version of the immediacy hypothesis; at least some of the process of retrieving the antecedent of a pronoun is not immediate (see Carroll & Slowiaczek, 1987, for evidence that strengthens the case against the immediacy hypothesis).

However, the distance effect may not really be an effect of distance in the surface structure (i.e., how many words apart the pronoun is from the antecedent). Linguistic analyses of discourse structure (e.g., Grosz, Joshi, & Weinstein, 1983, 1995) have suggested that structural factors, not simple distance, determine the appropriateness of using a pronoun to co-refer with an antecedent noun phrase. Clifton and Ferreira (1987) tested the possibility that distance would not matter as long as the antecedent noun was still the “topic of the discourse” and thus still in some sort of active memory (see also, Garrod & Sanford, 1983). They tested this hypothesis using a phrase reading task, in which the reader saw a passage a phrase at a time and pushed a button to get the next phrase to appear. Reading time for a phrase was measured by the time between two button pushes. An example of one of their passages is as follows:

Weddings can be / very emotional experiences for everyone involved. / The
cigar smoking caterer / was obviously / on the verge / of tears, / and the others
/ were pretty upset too. / In fact, / the organist, / who was an old maid, /
looked across the room / and sighed. / She was / still looking for / a husband. (9.5a)

(The slashes indicate the boundaries between phrases.) Of primary interest was the reading time for the underlined target sentence containing the pronoun *she*. In this example, *the organist* is the topic of the previous sentence and thus, according to this explanation, quite available as the antecedent for the pronoun *she* in the last sentence. On the other hand, if the middle two sentences are replaced by

The cigar smoking caterer / was obviously / on the verge of tears, / having
just noticed that / the organist, / who was an old maid, / was holding hands
/ with someone else. (9.5b)

the organist is not the topic of the sentence prior to the target sentence, and thus would not be available in active memory. Clifton and Ferreira found, in fact, that reading time for the sentence containing the pronoun was longer when the pronoun’s antecedent was no longer the topic than when it was. As with the reading experiments measuring eye movements, a large portion of the effect was delayed, appearing after the phrase containing the pronoun. Clifton and Ferreira also varied how far back in the passage that the antecedent *organist* was, and found that distance, per se, had no effect on reading time for the target sentence as long as topic status remained the same.

Research by Gordon and his colleagues (Gordon, Grosz, & Gilliom, 1993; Gordon & Hendrick, 1997, 1998) has used a more precise and detailed analysis of the structural factors that affect finding the antecedent of a pronoun. They based their work on an analysis of “centering” by Grosz, Joshi, and Weinstein (1983, 1995) that combines linguistic and artificial intelligence approaches to language. This analysis draws on the idea that in a coherent discourse, sentences tend to keep talking about entities and events that were mentioned in earlier sentences. Centering theory (Grosz et al., 1983, 1995) defines a set of forward-looking centers, which are the entities and events in a sentence most likely to be referenced in future sentences, and a backward-looking center, which is the entity or event that is referenced in the current sentence that points back to a high-ranked forward-looking center of the previous sentence. The backward-looking center must be realized as a pronoun, if any element in its sentence is.

Gordon and his colleagues examined several implications of a psycholinguistic elaboration of this analysis, in which they proposed that a backward-looking center should generally be realized as a pronoun. One notable finding was the repeated name penalty. They measured the time to read passages like 9.6:

Bruno was the bully of the neighborhood. He chased Tommy all the way home from school one day. He watched him hide behind a big tree and start to cry. (9.6a)

Bruno was the bully of the neighborhood. Bruno chased Tommy all the way home from school one day. Bruno watched Tommy hide behind a big tree and start to cry. (9.6b)

The later sentences in 9.6b were read more slowly than those in 9.6a. Gordon et al. proposed that this was because the pronoun in the second sentence of 9.6a is an appropriate backwards-looking center, suitable to be realized as a pronoun. Realizing it as a full noun phrase, as in 9.6b, indicates that it is not the backwards-looking center, disrupting discourse integration. (Note that the third sentence in 9.6a is read quickly as well, indicating that if the backwards looking center of that sentence—its subject—is realized as a pronoun, other phrases can be as well. Thus, readers show no disruption upon reading the next pronoun, *him*, in this sentence.)

Although a pronoun is efficiently understood when its antecedent is a high-ranking forward-looking center of a previous sentence, other factors determine how and how easily it is understood. Morphological marking (for gender and number in English) clearly plays an important role. Sentences like 9.7a are read faster than 9.7b, presumably because gender-marking makes the former unambiguous while the reader has to resolve the ambiguity of reference in the latter even though it is probably clear that *he* refers to Ron (Ehrlich, 1980):

Sally rewarded Ron because he was on time. (9.7a)

Sam rewarded Ron because he was on time. (9.7b)

Vonk (1984) examined gender ambiguity in combination with another factor that has been shown to affect pronoun comprehension, implicit causality (Caramazza, Grober, Garvey, & Yates, 1977). She measured eye movements while participants read sentences that began like those in 9.8 and immediately afterwards named the antecedent of the pronoun:

Mary won the money from John, because she ... (9.8a)

Mary won the money from Jane, because she ... (9.8b)

John punished Mary, because she ... (9.8c)

Jane punished Mary, because she ... (9.8d)

These sentences contain verbs that implicitly specify the causal agent responsible for an action. Mary is the responsible agent in all these sentences (just as Ron is in 9.7), and therefore the preferred antecedent for the pronoun in the *because* clause. Vonk found that participants named the correct antecedent virtually all the time, even in sentences like 9.8b and 9.8d, when the gender of

the pronoun did not disambiguate the assignment.

A very interesting finding emerged from the pattern of fixation times. Vonk distinguished between first-pass times (time spent fixating a region before any regressions from it) and second-pass times (time spent after a regression is made into the region; see Chapter 4). The first-pass times on the pronoun indicated that fixations were longer when the gender information made assignment unambiguous (i.e., when the gender of the two names was different) than when it didn't. In contrast, first-pass fixation times on the verb phrase following the pronoun as well as second-pass fixation times on the entire sentence were longer when only the implicit causality of the verb disambiguated the assignment. It appears that readers quickly assign the antecedent of a pronoun that is disambiguated by gender, but delay making the assignment when it is determined by implicit causality.

In the sentences that Vonk used, the pronoun could always co-refer to an appropriate antecedent. Koornneef and van Berkum (2006) measured reading time (using both self-paced reading and eye tracking) while people read sentences where the pronoun was either consistent or inconsistent with the implicit causality bias, such as the Dutch versions of those in 9.9:

Linda praised David because he had been able to complete the difficult assignment
with very little help. (9.9a)

David praised Linda because he had been able to complete the difficult assignment
with her help only. (9.9b)

The verb *praised* implicitly indicates that its direct object, the recipient of the praising, is the likely causal agent for a following *because* clause, and thus the preferred antecedent of the pronoun subject of that clause. Sentences like 9.9a, in which the gender of the pronoun was consistent with this assignment, were read faster than inconsistent sentences like 9.9b.

Many other factors have been found to affect the comprehension of pronouns. For instance, readers find it relatively difficult to “go inside” a conjoined noun phrase to find an antecedent for a singular pronoun (e.g., to refer to Mary as *she* following a phrase like *John and Mary went into town*; Albrecht & Clifton, 1998), although this difficulty does seem to depend on some subtle properties of the act that the conjoined pair of individuals are engaged in (Koh, Sanford, Clifton, & Dawydiak, 2008). Also, participants are faster to read a sentence containing a pronoun when its antecedent was introduced as a proper name as opposed to a descriptive noun phrase (Sanford, Moar, & Garrod, 1988). But we must turn to a discussion of other factors that affect the coherence of a text.

Nominal anaphora

Noun phrases as well as pronouns can refer to previously introduced referents. An indefinite noun phrase typically introduces a new referent into discourse, while a definite noun phrase frequently refers to a previously mentioned entity, especially when the referent is not salient enough for a pronoun to refer to it (cf. Almor, 1999). In contrast to pronouns, a definite noun phrase can refer to an entity that a reader could be expected to have inferred, as *the seat* in 9.10, or that the reader will recognize as unique, as *the biggest supermarket* in 9.11. (In passing, we have to say that the semantics of definiteness are actually quite complicated—try telling a speaker of languages that do not have definite vs. definite articles, e.g., Chinese, Japanese, or Korean, exactly when it is appropriate to use the definite article *the* and when it is not. See Heim & Kratzer, 1998, for discussion of the grammar of definiteness.)

Joe jumped into a nearby cab. The seat was disgustingly dirty. (9.10)

Mary looked for shallots in her local grocery store. Eventually she found them in the biggest supermarket in town. (9.11)

Despite these complexities, readers are exquisitely sensitive to whether a definite noun phrase has an appropriate antecedent in a discourse. Yekovich and Walker (1978) investigated two-sentence passages like 9.12a and 9.12b:

The accused youth sobbed quietly and asked for (the/an) attorney.
The attorney examined the statement of the arresting officer. (9.12a)

The contented mother rocked slowly and hummed (the/a) tune. The tune soothed the temper of the fussy newborn. (9.12b)

The second sentence of 9.12a was read more slowly when the first sentence had contained *an attorney* than when it had contained *the attorney*. Presumably, this is because simply asking for *an attorney* doesn't introduce some specific attorney into the discourse (although referring to *the attorney* in the first sentence does induce the reader to accommodate the existence of one, who can later be referred to with a definite noun phrase). However, no comparable difference was observed in 9.12b. This is most likely because *humming a tune*, unlike *asking for an attorney*, does introduce some specific tune into the discourse, providing *the tune* with a referent.

Quite often a superordinate noun phrase is appropriately used to refer to an entity previously introduced by a more specific noun phrase. Consider materials studied by Garrod and Sanford (1977), illustrated in 9.13:

A robin would sometimes wander into the house. The bird was attracted by the larder. (9.13a)

A goose would sometimes wander into the house. The bird was attracted by the larder. (9.13b)

They measured the reading time for each sentence. They found that readers took longer to read the second sentence in examples like 9.13b than 9.13a. If the concept in the first sentence is a less-typical member of the superordinate category, it is apparently accessed as a referent of the superordinate term *the bird* less quickly. But typicality only matters when the term in question actually provides the antecedent for the superordinate term. The second sentence in 9.14 was read no more quickly when the typical vehicle (a bus) was mentioned in the first sentence than when the less-typical vehicle was. The effect seen in 9.13 is not a simple matter of a more typical term priming its category more than a less-typical term.

A (bus,tank) came roaring round the corner.
It nearly hit a horse-drawn vehicle. (9.14)

Relations between propositions

Anaphora is not the only linguistic device that holds a text together. A readable text flows well from one proposition to the next. When the relation (causal, temporal, or otherwise) between

two propositions is not obvious, a text can provide linguistic “connectives,” like *because* and *so* and *but* and *however* that guide the reader in putting the sentences together (Halliday & Hasan, 1976).

Readers are sensitive to these relations and how they are signaled. Reading is slowed when the temporal order of propositions in a text does not match the chronological order of the events they denote (Mandler, 1986). Reading is also slower when there is a large time shift in the narrative than when events follow directly upon one another (Zwaan, 1996). For instance, 9.15a is read more slowly than 9.15b:

Jamie turned on his PC and started typing. An hour later, the telephone rang. (9.15a)

Jamie turned on his PC and started typing. A moment later, the telephone rang. (9.15b)

Thematic coherence matters too. Imagine you are reading a narrative about someone who is in a kitchen preparing dinner. In the middle of this narrative there is a sentence like either 9.16a or 9.16b:

I cut up a slice of cooked ham. (9.16a)

I took my moped from the garage. (9.16b)

Not too surprisingly the thematically incongruent sentence, 9.16b, is read more slowly than the congruent one, 9.16a (Bestgen & Vonk, 2000). Very interestingly, however, if the target sentence was preceded by a temporal adverb that indicates a change in time and possible setting (e.g., *Around 11 o'clock*), the reading time cost of the thematically incongruent sentence becomes smaller and non-significant.

Linguistic connectives are designed to guide a reader in identifying the relations among sentences, and readers do use them. Haberlandt (1982), for instance, had people read passages with causal or adversative connectives, such as 9.17:

The passengers were terrified. They thought the plane would crash. (However), the pilot made a safe landing. (9.17)

Reading was faster when the connective (the adversative *however* in this example) was present than when it was not. Apparently, the connective speeded the task of determining how the last two sentences are related.

Information structure and the “given–new contract”

The syntax and morphology of a language provide ways to unambiguously encode the thematic roles that phrases play in sentences: “Who did what to whom.” Languages also provide a variety of devices for linking sentences together in discourses. We have discussed anaphora as one such device, and the provision of connectives as another. Languages also have devices to encode the “information structure” of sentences. These devices signal distinctions that go by names like topic vs. comment, ground vs. focus, theme vs. rheme, given vs. new, old vs. new, etc. (see Vallduví & Engdahl, 1996, for a thorough discussion of these contrasts and how they are realized in different languages). These linguistic devices can all be seen as ways that a writer can aid a reader in relating

the propositions in one sentence to the representation the reader has constructed for the preceding text (Clark & Haviland, 1977). They signal what information in a sentence should be considered new and informative, and what information the writer presupposes that the reader already has in the mental representation of the discourse.

These devices take different forms in different languages. In spoken English the most important devices are prosodic: New and focused information has to receive a “pitch accent” (generally a rise in pitch, sometimes accompanied by an increase in duration and/or amplitude; Selkirk, 2003). Other languages have specific syntactic structures for signaling what is given or topic and what is new or focus (Vallduví & Engdahl, 1996). Even English has syntactic ways of encoding information structure. By default, a topic appears as the subject of a sentence, and a focused phrase appears as the final element of its clause. Focus can also be signaled by specific devices such as clefting; in a sentence like *It was the whiskey that made Tom sick*, the phrase *the whiskey* is focused, and it is presupposed that the reader knows that Tom was sick (in contrast to *What the whiskey did was make Tom sick*).

Psychologists have most often studied information structure from the standpoint of given vs. new information. Given information is information that the reader can be presupposed to already have in his or her mental model of the discourse. New information is information that is supposed to be added to the mental model, linked to the given information. Clark and Haviland (1977) proposed that there is an implicit “contract” between writer and reader (or speaker and hearer) to phrase sentences in a way that makes it clear what links to the prior discourse and what adds new information. Perhaps the clearest case of how this contract is realized can be seen in question–answer pairs. The pair in 9.18a respects the given–new contract. Tom is the topic or theme or given information in the answer, and *Tom* appears as the subject. *Whisky* is the focus or rheme or new information, and it appears as the predicate. But the pair in 9.18b violates the given–new contract. The information that is given in the text appears in the clefted position, where new information should appear, and the new information appears in the position that should contain given or presupposed information:

Q: What did Tom drink? A: Tom drank WHISKY. (9.18a)

Q: What did Tom drink? A: It was TOM who drank whisky. (9.18b)

The effects of such violations of information structure have most often been studied in spoken English (largely because, as mentioned earlier, the primary device English has for signaling focus is prosody). Violations of information structure (as in 9.18b) have been shown to impair auditory comprehension (Bock & Mazella, 1983; Birch & Clifton, 1995). Hornby (1974) showed that listeners seemed to pay less attention to material that was expressed as if it were given than if it were expressed as new information. For instance, if they had to decide whether or not a sentence they heard accurately described a briefly presented picture of a girl petting a cat, they were more likely to overlook the mismatch between the sentence and the picture if they had heard 9.19a than if they had heard 9.19b:

It is the girl that is petting the dog. (9.19a)

It is the boy that is petting the cat. (9.19b)

The difference is that in the former sentence, the critical information (about petting the dog) was presented as old, given information. In the latter sentence the critical information (about the

boy doing the petting) was presented as new, focused information. A practical implication is that if you want to put something over on your reader without arousing the reader's critical faculty, express it as given information.

Reading experiments also show that focused information receives more attention. Birch, Albrecht, and Myers (2000) showed that recognition probes that were delayed 10 seconds after the end of a text were responded to more quickly when the word they probed had appeared in a focused position (e.g., an it-cleft, such as *mayor* in *It was the mayor who refused to answer a reporter's question* than if *mayor* had appeared in an unmarked subject position). In a reading time experiment Birch and Rayner (1997) demonstrated that readers fixated focused information longer than given information. In one experiment they measured eye movements while subjects read passages like 9.20, which began with one version of a focus-inducing question (Q-A or Q-B) and continued with an answer:

Q-A. Where were the soldiers?

Q-B. What were the soldiers playing?

A. The soldiers in the underground bunker were playing cards to relieve their boredom. (9.20)

Following Q-A, the focused information was the phrase *in the underground bunker* (wide focus). Following Q-B, it was the single word *cards* (narrow focus). Gaze duration for the narrowly-focused word *cards* was not affected by the focus manipulation, although the word was re-read for a longer time when focused by Q-B. First pass time for the phrase that was the target of wide focus in Q-A (*in the underground bunker*) was increased by the focus. These data suggest that focused material receives more processing, consistent with an ERP result to be discussed soon. However, Birch and Rayner (2010) found faster reading time for material that they described as more "prominent" in the discourse, including material that would be considered to be focused, so the actual relation between focus, prominence, attention, and reading speed remains somewhat uncertain.

The information structure of a sentence ideally reflects how the sentence relates to the discourse structure, but it can also affect how a sentence is interpreted. Consider the Late Closure parsing preference discussed in the previous chapter. Sentences like 9.21a are read faster than sentences like 9.21b, because the former permits the final adverb phrase to be attached to the current clause (*she proposed . . .*) while the latter's tense forces it to modify the earlier clause (*Fiona will implement . . .*):

Fiona will implement the plan she proposed to the committee last week. (9.21a)

Fiona will implement the plan she proposed to the committee next week. (9.21b)

However, when Altmann, van Nice, Garnham, and Henstra (1998) had people read these sentences in a context like 9.22, the advantage of the late closure version (*last week*) disappears and is replaced with an apparent early closure (*next week*) advantage in several eye-tracking measures. What is going on here is that the discourse contains an indirect question (*The other committee members wonder when . . .*) about the time that the plan will be implemented, making the answer to this question (*She'll implement the plan . . . next week*) the focus of the final sentence. This focus seems to be sufficient to resolve the ambiguity of how the final adverb phrase will be attached. (Note, though, that to obtain this result, Altmann et al. had to insert a phrase *to the committee* between the adverb phrase and its most recent attachment site, which eliminated the normal late closure parsing preference.)

Last week, Fiona presented a new funding plan to her church committee. The other committee members wonder when Fiona will implement the plan she proposed. She'll implement the plan she proposed to the committee last week/next week, they hope. (9.22)

In an event-related potential (ERP) experiment Cowles, Kluender, Kutas, and Polinsky (2007) showed distinctive scalp electrical potential responses to information that appeared in a focused position. Their subjects read short passages like 9.23:

A queen, and advisor, and a banker were arguing over taxes. Who did the queen silence with a word, the advisor or a banker? It was the banker that the queen silenced. (9.23)

Cowles et al. showed that a word that appears in a focused position (e.g., the *it*-cleft, or even the final position of a sentence) elicited a brain electrical response (a P3b) that signals increased processing of the word. This is presumably related to Hornby's (1974) demonstration that listeners are more sensitive to focused than to presupposed information and to Birch et al.'s (2000) demonstration that probe responses can be faster to focused than non-focused information.

Not only do readers seem to pay more attention to new and focused information, they also seem to be disrupted, at least briefly, either if information that has not been presented in the preceding text appears as given information or if information that is given in the preceding text appears in a focused position. Haviland and Clark (1974) demonstrated the former by measuring the time to read the second sentence in short passages like 9.24:

Horace got some beer out of the trunk. The beer was warm. (9.24a)

Horace got some picnic supplies out of the trunk. The beer was warm. (9.24b)

Horace was especially fond of beer. The beer was warm. (9.24c)

In all cases, the phrase *the beer* appears as a definite noun phrase in the subject position of the second sentence, a signal to the reader that the phrase refers to given information. This is the case in 9.24a. However, no beer had been introduced in 9.24b. The reader has to make an inference that beer must have been included in the picnic supplies and add beer to the discourse representation. This takes extra time, slowing the reading of the second sentence of 9.24b compared to 9.24a. Similarly slowed reading was observed in 9.24c, where the word *beer* had appeared previously, but in a way that does not actually introduce a referent for *the beer*.

Cowles et al.'s (2007) study provides evidence that reading given information in a focused position can be disruptive. In the passage in 9.23, the answer *It was the banker that the queen silenced* has an appropriate information structure. The answer to the question, *the banker*, which should be focused, appears in the focused position in the *it*-cleft. However, in other passages, the information structure was inappropriate. Given information was presented in the focused position, e.g., *It was the queen that silenced the banker*. When such inappropriate information appeared in focused position, it elicited an N400 brain electrical response. The N400 is typically seen when an unexpected or semantically inappropriate word is read, suggesting that the readers took focusing a given term to be inappropriate.

Reinstatement from memory

We have seen how texts can provide a variety of aids to help a reader put information together, and how readers respond to such aids. But sometimes a text presents a reader with other challenges. For instance, a reader may be required to relate new information to some information that was presented so far back in the text that it is no longer easily available. You may have had the experience of reading a newspaper article and seeing a person's name mentioned, but not remembering who in the world that person was. You have to search the preceding text to recover that information. But there are circumstances in which material you read "automatically" calls to mind material to which it is related, even though that material occurred much earlier and is no longer in an active memory.

Some researchers propose that this reinstatement of earlier material occurs because of a passive "resonance" process (McKoon, Gerrig, & Greene, 1996; Myers & O'Brien, 1998). Such a process can account for data provided by Albrecht and O'Brien (1993), whose participants read long passages in which (for instance) the main character, Mary, was introduced as a strict vegetarian. After several filler sentences, which did not talk about Mary or her vegetarianism, the reader encountered a sentence that described Mary as ordering a cheeseburger and fries. This sentence was perfectly coherent with the preceding few sentences, but nonetheless was read more slowly than it was in a neutral passage that had given no information about Mary's eating habits. Albrecht and O'Brien proposed that the proposition about ordering a cheeseburger reactivated the information about Mary being a vegetarian through memory a resonance process, allowing the reader to detect the inconsistency in the text. Later research (O'Brien, Rizella, Albrecht, & Halleran, 1998) indicated that this resonance process is not particularly intelligent. A similar slowing of reading was observed even though the text that intervened between the mention of Mary's vegetarianism and the ordering of the hamburger explicitly said that Mary was no longer a vegetarian. (See O'Brien, Cook, & Guéraud, 2010, for further evidence confirming that information a reader knows is outdated still affects text comprehension.)

Other researchers grant readers more intelligence, and emphasize how the reader's search for meaning can hold a text together (Graesser, Singer, & Trabasso, 1994). Goals and causal actions play particularly important roles in this process. Suh and Trabasso (1993), for example, have shown that, in a text describing how a child wanted a new bicycle, memory for the bicycle goal was more salient if the goal was not yet satisfied than if it had been satisfied. Very likely both intelligent, meaning-guided, and rather dumb, resonance processes are involved in how memory helps glue texts together (Myers & O'Brien, 1998). Albrecht and Myers (1995) measured the time to read sentences in a story in which the protagonist was described as being successful or not in making an airline reservation late at night (thus satisfying or not satisfying a goal), followed by a target sentence that was inconsistent with the earlier goal not having been satisfied (stating that the protagonist was tired and decided to go to bed). When six or seven sentences that did not involve the goal followed the text about the goal, the target sentence was read equally fast regardless of whether or not the goal had been satisfied. Apparently, readers did not notice the inconsistency between not having made the airline reservations and going to bed. However, if some event was arbitrarily associated with making the airline reservation (e.g., sitting on a leather couch to look up the airline's phone number), calling this event to mind (by referring to the leather couch immediately before the target sentence) was enough to slow readers down on the target sentence if the goal of making an airline reservation had not been satisfied. Apparently, the arbitrary connection between the leather couch and the goal of making the reservation was enough to resonate with the goal statement, reminding the reader that the reservation had or had not been made. In sum, it appears that both passive resonance processes and active processes of meaningful interpretation play roles in text comprehension.

Inferences

A reader often has to do more than tie the sentences of a text together. The reader has to go beyond the information provided and add content that is only implicit in the text (see Singer, 2007, for an overview of inference processing in discourses). Much of the material earlier in this chapter has actually touched on inference processes. Deciding what the antecedent of a pronoun is can require inferences (e.g., in the fragment *Joe praised Tom because he . . .*, the referent of *he* is probably *Tom*, the person who did something that deserved praise; but this is an inference). Adding an entity to a discourse model to support a definite reference (e.g., *The beer was warm*) similarly requires going beyond the information presented, if the beer had not been mentioned before. Deciding that the protagonist of a story had not satisfied her goal of making an airline reservation because something had distracted her when she started to do it likewise requires an inference.

Making an inference takes time, and slows a reader down. An experiment reported by Haviland and Clark (1974) demonstrated that the second sentence of 9.25a:

Last Christmas Emily went to a lot of parties.
This Christmas she got very drunk again. (9.25a)

takes longer to read than the same sentence in the context 9.25b

Last Christmas Emily became absolutely smashed.
This Christmas she got very drunk again. (9.25b)

The latter text provides an explicit antecedent for *got very drunk again*, but the antecedent has to be inferred in the former text. The inference must not have always been made while reading the first sentence, because if it had been made, the second sentence should have been read equally fast in both contexts.

This example brings up the question of what inferences are made, and when. It is clear that a reader cannot make all possible inferences, or even the likely ones. There are just too many. A general conclusion from a lot of research is that readers are cautious. They often appear to wait until they must make an inference before they make it. But the detailed answer to what inferences are made, and when, appears to differ for different types of inferences. We will discuss two types of inferences: bridging inferences and elaborative inferences. A warning: Different experimental techniques seem to yield somewhat different results. The usual techniques that have been used to study the inferencing process include measuring the time to read a sentence whose interpretation requires that an inference is made (including measuring eye movements), the time to make a lexical decision or a naming response to a word that is presumably activated by an inference, the time to recall a sentence in response to a cue word that was presumably inferred, the time to verify the truth of a sentence that could have been inferred. When the results of different techniques all suggest that an inference has been made, one can have confidence in that conclusion. But quite often, techniques point in different directions, making conclusions uncertain.

Bridging inferences

A bridging inference is made when something has to be added to the discourse model to tie a new proposition into the previous model. Without such an inference the text would lack coherence. Anaphoric inference is an example of a bridging inference; a reader has to figure out who *he* refers to in *John praised Bill because he caught the pass* in order to make sense of that

sentence. For another example, our discussion of Haviland and Clark (1974) presented the classic example of a bridging inference. You will recall that Haviland and Clark had participants read passages like 9.24 (repeated below) and found that the second sentence was read more slowly in 9.24b, where the reader had to infer that the picnic supplies contained beer, than in 9.24a:

Horace got some beer out of the trunk. The beer was warm. (9.24a)

Horace got some picnic supplies out of the trunk. The beer was warm. (9.24b)

The reader has to “bridge” the gap between *picnic supplies* and *the beer* to know what *beer* probably refers to. This requires the reader to use his or her background knowledge that beer is plausibly included in the picnic supplies. But note that the gap is not bridged until it must be; the reader generally does not infer that beer (and sausages and chips and ice and napkins . . .) is included in the picnic supplies until the inference is needed.

Another example of testing whether bridging inferences are made, perhaps after a delay, comes from Keenan and Kintsch (1974). They had people read passages like *Police are hunting a man in hiding. The wife of Bob Birch disclosed illegal business processes in an interview on Sunday.* One way of making sense of this passage is to infer that Bob Birch is the man in hiding. Readers apparently did this, at least by the time a verification test (“Was Bob Birch the man in hiding?”) was given 15 minutes after reading the text. They were as fast verifying a statement like this as when it had been explicitly presented in the text.

Bridging inferences can involve goal-related and causal relations as well as identifying entities in the discourse. Myers, Shinjo, and Duffy (1987) had people read the sentence in 9.26b after reading one of the sentences in 9.26a:

High relatedness: Tony’s friend suddenly pushed him into a pond.

Medium relatedness: Tony met his friend near a pond in the park.

Low relatedness: Tony sat under a tree reading a good book. (9.26a)

Test sentence: He walked home, soaking wet, to change his clothes. (9.26b)

The test sentence was read more slowly as the preceding sentence was less related to it. Presumably, the extra time was spent in trying to inferentially bridge the relation between the two sentences. Interestingly, when knowledge for how Tony got wet was later probed, recall was better in the medium relatedness passage than in either of the others. Apparently, the extra work done to bridge the two sentences enhanced memory, but only when it is successful (which was apparently not the case in the low relatedness passage).

A final example of a causal bridging inference comes from Singer and Halldorson (1996), who had participants read passages like 9.27:

Terry was unhappy with his dental health.

He phoned the dentist. (9.27a)

Terry was unhappy with his dental bill.

He phoned the dentist. (9.27b)

Three seconds after reading the second sentence, the participant was presented with a sentence like *Do dentists require appointments?* and asked to say whether it was true or false in general.

The decision was made more quickly following the passage in 9.27a than following the similar one in 9.27b. Apparently readers found a way to make a causal bridge between the two sentences. When the first concerned dental health, the inference that is made is that Terry called the dentist to make an appointment. However, when the first sentence concerned dental bills, the inference is that Terry called the dentist to complain. Making the bridge in the former case activated pre-existing knowledge that dentists require appointments, which in turn speeded verification.

Elaborative inferences

Sometimes inferences are made even before they are needed to make a text coherent. As an example, Garrod and Sanford (1982) had people read one of the sentences in 9.28a followed by 9.28b:

Explicit: Keith took his car to London.

Implicit: Keith drove to London. (9.28a)

Test: The car kept overheating. (9.28b)

The test sentence was read equally fast in both cases. This contrasts with the Clark and Haviland case (*The beer was warm*) discussed earlier. Apparently, the discourse representation constructed in the implicit version of 9.28a included a car (while the picnic supplies did not necessarily include beer). Still, since no car was explicitly mentioned in the text, its existence required an inference—and since its existence was not needed to make the sentence *Keith drove to London* coherent, it could not be considered a bridging inference. It is generally called an elaborative inference.

Such elaborative inferences are not always made. As indicated before, there may just be too many of them. But it is of interest to examine the conditions under which they are made. Consider an example discussed earlier, from Garrod and Sanford (1977), and shown in 9.13 (reproduced again below):

A robin/goose would sometimes wander into the house. The bird was attracted by the larder. (9.13)

The reading time for the second sentence was longer when the preceding sentence contained the atypical *goose* than when it contained the typical *robin*, suggesting that the typical term was more likely to trigger an inference to the superordinate concept of bird. In addition, reversing the order of the specific and general terms, as in 9.29, increased reading time for the second sentence.

A bird would sometimes wander into the house. The robin/goose was attracted by the larder. (9.29)

It appears that reading *robin* or *goose* in 9.29 requires that the reader add some specific information to the representation of the text, slowing reading compared to 9.13. Reading a word denoting a specific concept may not generally activate a specific example of the category, even a highly typical one (although reading was faster for *robin* than for *goose* in 9.29, suggesting that the inference that it is the bird previously referred to was easier for the more typical instance).

However, if the context is constraining enough, readers may make an elaborative inference from a general category to a specific example (Garrod, O'Brien, Morris, & Rayner, 1990). Participants

read *The robin pecked the ground* about equally fast after 9.30a, where robin is only implicit, as after 9.30b, where it is explicitly mentioned:

Julie was convinced that spring was near when she saw a cute red-breasted bird in her yard. (9.30a)

Julie was convinced that spring was near when she saw a cute red-breasted robin in her yard. (9.30b)

Early work on elaborative inferences did suggest that such inferences were quite frequently drawn. Much of this work relied on assessments of memory for text. For instance, Johnson, Bransford, and Soloman (1973) gave participants passages like *The man dropped the delicate glass pitcher on the floor*. Later, they were about as likely to recognize an implication of this sentence, *The man broke the delicate glass pitcher on the floor*, as the actual sentence. However, later work (e.g., McKoon & Ratcliff, 1986) suggested that such memory effects did not demand the conclusion that the elaborative inference had been made when the passage had been presented. Instead, this later work suggested that the memory test could, instead, provide an effective retrieval cue when it expressed a predictable outcome. A similar conclusion was suggested by Corbett and Doshier (1978) in the domain of instrumental inferences (inferences made from a verb to a typical instrument) (see also Alba & Hasher, 1983).

Singer (1979) provided evidence from reading time measures that similarly indicated that instrumental inferences are not necessarily made at the time of reading. His participants read pairs of sentences such as 9.31a and 9.31b:

The boy cleared the snow with a shovel. The shovel was heavy. (9.31a)

The boy cleared the snow from the stairs. The shovel was heavy. (9.31b)

The reading time for the second sentence was longer in 9.31b than in 9.31a, indicating that integrating *shovel* with the prior sentence was more difficult when it was not explicitly mentioned. Singer reasoned that if readers had inferred *shovel* while reading the first sentence of 9.31b, then there should have been no difference.

However, it may be that instrumental inferences, like category-example inferences, are made only when they are highly constrained. Consider the passage in 9.32, from O'Brien, Shank, Myers, and Rayner (1988):

All the mugger wanted was to steal the woman's money. But when she screamed, he stabbed her with his weapon in an attempt to quiet her. He looked to see if anyone had seen him. He threw the knife into the bushes, took her money, and ran away. (9.32)

The inference from *he stabbed her with his weapon* to the conclusion that the weapon was a knife is not a logical inference like knowing that a robin is a bird: the weapon could be an icepick, spear, or any other sharp instrument. However, O'Brien et al. measured eye movements while reading the last sentence of 9.32 and found fixation durations on *knife* were no longer than when the sentence followed a version of the passage in which *knife* had been explicitly mentioned previously (i.e., *weapon* was replaced with *knife*). In contrast, when the verb *stabbed* in the second sentence was replaced by a verb that did not strongly suggest a particular instrument (e.g., *assaulted*), gaze dura-

tion on *knife* was increased. It appears that readers do not always draw elaborative inferences when the context is unclear, and that the search for an antecedent of a definite noun phrase like *the knife* is begun immediately. Still, when a verb has a highly typical instrument, like the *stab-knife* pair, the instrumental inference may be made.

A different type of elaborative inference involves the causal relations in the situation a text describes. We have already discussed (in connection with 9.26 and 9.27) how causal inferences are made to bridge between two superficially unrelated sentences in a text. But we can also ask whether likely causal relations are inferred before they are required for discourse coherence. Consider passages like 9.33, as studied by Duffy (1986):

John was eating in the dining car of a train. The waiter brought him a bowl of soup. Suddenly the train screeched to a halt. The soup spilled in John's lap. (9.33)

After finishing this passage, careful readers undoubtedly make a causal link between the sudden halting of the train and the soup spilling. However, do they infer that the soup will spill in John's lap before they get to the last sentence? This type of inference is not logically required. The soup might not spill if the bowl isn't very full and it would only spill in John's lap if he were sitting on the forward side of the table. Yet the event of the soup spilling seems probable, and furthermore the narrative seems to be pointing in that direction.

Duffy (1986) distinguished between three possibilities of how such a causal link might be drawn. The first is backward inference or bridging: the reader waits until getting to the soup spilling sentence before searching back in the text for the cause or explanation of the event. The second is specific expectation, namely that the reader generates a specific expectation or prediction from the first three sentences that the soup is going to spill that is confirmed by the fourth sentence. The third is focusing, in which certain aspects of the text are so salient that they advertise themselves as things that are likely to be followed up on. When something comes up which has no transparent precedent, these salient items in memory are assumed to be the appropriate places to find the rationale for what follows. One might view this salience model as the reader using mental highlighter on some sentences in memory.

In one experiment Duffy presented participants with passages like the one above. For example, they read the first three sentences of 9.33 and then were shown the fourth sentence and asked to judge whether it was a good continuation of the passage. The control passage had a different third sentence, *The train began to slow down entering a station*, which makes the causal link less likely but still possible. Unsurprisingly, readers took less time in judging that the fourth sentence was an appropriate continuation of the first (*screeching to a halt*) version of the passage than of the second (*slow down*) version, indicating that they were making a causal inference at some point.

What was of greater interest was that when the fourth sentence was not a good continuation, such as *That night the whole forest burned down*, readers were quicker to judge that it was not a good continuation when the third sentence was the *screech to a halt* sentence than when it was the *slow down* sentence. This appears to rule out the backward inference model. If readers made such causal inferences merely searching back from the last sentence, then the time to judge that the "fire" sentence was not a good continuation should have been the same since there was nothing about fires in either version of the passage.

The difference indicates that readers are doing something active to the text that is shaping how they will process the next sentence they encounter. But are they actually generating a specific prediction of what is coming up next? Duffy assessed this by using a probe memory test. Participants read passages like the "soup" one and then, after either the *screeched to a stop* or the *slow*

down sentence, they got a single probe word (*soup*) and judged whether it had been in the passage or not. The idea is that if the *screeched to a stop* sentence generated a specific prediction of soup spilling, *soup* should have been more available and hence easier to judge than it had been in the passage. In fact, there was no difference between the two conditions, suggesting that such an inference was not made. More recent work has suggested that while highly specific inferences (such as Duffy's inference about soup-spilling) are made only in highly constraining contexts, more general inferences involving categories of words or features of such categories are made in less-constraining contexts (Harmon-Vukic, Guéraud, Lassonde, & O'Brien, 2009; Lassonde & O'Brien, 2009).

Duffy's data thus suggest that the process of causal inference in reading is complex. It is neither sitting back and waiting until you don't understand why something is true and then searching, nor is it making uncertain guesses about what is going to happen as you go along. Instead, it appears to be a process of highlighting information that you are likely to need later on so that the highlighted information is readily accessible for future linkage. What indicates to the reader that the *screeching train* sentence should be highlighted? In this case it seems to be important because the topic apparently has been changed from the meal to the train even though there has been no closure on the eating episode. Thus the reader seems to be warned that the *screeching train* will be important, somehow.

A final type of inference is one that can be made by combining two or more propositions in the discourse, or one or more discourse propositions together with background knowledge, and deducing an implication. If readers do make the inference, reading of a later proposition in the text that contradicts it should be disrupted. Researchers have quite often failed to detect such disruption, but sometimes disruption is observed—e.g., when the subject matter is quite familiar to the reader or if the inference is specifically relevant to the goal of reading (see, e.g., Noordman, Vonk, & Kempff, 1992).

Another situation in which such deductive inferences are made is when all relevant information is presented in the text, in close enough contiguity to all be active in memory at one time. Wiley and Myers (2002) measured whole-sentence reading time of texts like 9.34a followed by a consistent (9.34b) or an inconsistent (9.34c) sentence:

Seals are usually found in cold regions. Like most animals in such regions, they usually have to produce a lot of energy just to keep warm. Metabolic rate increases with energy needs. (9.34a)

Seals have high metabolic rates. (9.34b)

Seals have low metabolic rates. (9.34c)

The target sentence was read more slowly when it was inconsistent with a presumed implication of the preceding text (9.34c) than when it was consistent (9.34b). While this experiment does not conclusively show when the inference was drawn, it does show that it was made at least by the time the critical sentence was read.

To summarize, readers generally appear to be cautious about making inferences when they read (Corbett & Doshier, 1978; Duffy, 1986; McKoon & Ratcliff, 1986; Singer, 1979; Singer & Ferreira, 1983). Even such obvious elaborative inferences as instrumental inferences do not appear to be made much of the time. Instead, readers only appear to make inferences when the inference is highly constrained, or when the inference is needed to make the current text tie in with what has come before. Why are readers so cautious? One possibility is that reading is demanding enough of processing resources even without elaborative inferences; there are words to be encoded, sentences

to be parsed, anaphoric antecedents to be searched, etc. A second possibility is that the costs of being wrong outweigh the benefits of being right. Many inferences that seem obvious when you know the answer are not in fact so obvious: the weapon didn't have to be a knife, the next sentence didn't have to be about soup spilling, it could have been about an accident or about robbers closing in on the train, etc. It seems likely that most real text is constructed so that there are very few sentences that merely confirm inferences that the reader has already made. Rather, as the given-new contract suggests, each sentence adds a new idea that the reader must tie in with what has gone on before.

Given the minimal effort view in the above paragraph, why does the reader bother to process certain passages more extensively ("highlight" them)? The answer may be that it is a good compromise between doing nothing and having to generate specific hypotheses. It may take little effort, and the hypothesis that something is likely to be used later on will be right much more often than a specific hypothesis about how it will be used. Clearly, much research is needed on what causes something to be made salient and targeted for future use as well as how backward search processes are influenced by what is salient. Duffy's (1986) experiment indicates that causal inferences make use of such highlighting, and our earlier discussion of centering and focusing indicates that potentially related processes play a variety of roles in helping a reader understand text.

Situation models

Many theorists argue that constructing a mental representation of the situation or scenario that is described is the principal goal of reading an expository text. Such a mental representation is often referred to as a "mental model" (Johnson-Laird, 1983); very similar concepts are termed situation models or discourse models (we will generally use the term situation model, but speak of mental model when we wish to emphasize the mental processes taking place). According to this view, a reader normally puts the sentences in an expository or narrative text together and makes at least the inferences needed for coherence with one primary purpose in mind: Constructing a situation model of what the text is about.

A situation model can represent a variety of aspects of a situation or event or scenario. It can represent the agents and objects that play roles in the situation. It can represent spatial relations among these agents and objects and it can represent visual and other sensory attributes of the situation. It can represent a situation from different perspectives. It can represent the motivations and the goals of the protagonists, and it can represent the causal relations between events that occur in the scenario. It can represent how a situation changes over time. A situation model, it seems, can represent anything we can know about a situation (see Zwaan & Radvansky, 1998, for one taxonomy of the types of information that situation models can represent).

A situation model is built up from the propositions contained in a text but it is both more and less than these propositions and the relations among them. It need not contain the details of how a situation was described, but it should represent the situation accurately. A compelling demonstration of how a mental model of a discourse goes beyond the propositions in the discourse is given by Sanford and Garrod (1998). If you read the short discourse *Harry put the wallpaper on the table. Then he put his mug of coffee on the paper*, everything seems fine. You understand the sentences, perhaps you create propositions like [ON, WALLPAPER, TABLE] and [ON, MUG, PAPER], and you interpret *paper* as anaphorically referring to the wallpaper. But now read *Harry put the wallpaper on the wall. Then he put his mug of coffee on the paper*. That does not compute! The propositions are nearly the same, but, because of background knowledge you bring to bear in constructing a representation of the situation being talked about, you know that the situation is impossible.

One way to experimentally study the role that memory models play in discourse comprehension is to study memory for texts. Consider an experiment by Garnham (1987), who had his participants read passages containing sentences like 9.35 (abbreviated from the original):

The party had been in progress for about four hours. . . . By the window was a man with a martini. He commented on the decor to a woman. . . . She asked him if he had complemented the host on his taste. The man standing by the window shouted to the host. The other guests looked toward the pair. . . . [and so on]. (9.35)

Later, after an intervening task, the participants were asked whether the text contained the sentence *The man with the martini shouted to the host* or the sentence *The man standing by the window shouted to the host*. If the participants had been warned that there would be a verbatim memory test, they were generally accurate. However, if they had not been so warned they were almost equally likely to choose either of these two sentences. However, they were very accurate in rejecting other candidate sentences such as *The host shouted to the man standing by the window*. They had apparently formed an accurate mental model of what happened at the party, but lost most of the information about how the text described the events.

The operation of mental models can be observed during or shortly after the reading of a paragraph. Glenberg, Meyer, and Lindem (1987) had participants read passages like that in 9.36:

John was preparing for a marathon in August. After doing a few warm-up exercises, he (took off/ put on) his sweatshirt and went jogging. He jogged halfway around the lake without too much difficulty. Further along the route, however, his muscles began to ache. (9.36)

On key trials participants would be probed after reading the paragraph on whether the word *sweatshirt* had been in it. Glenberg et al. found that participants took longer to respond “yes” in a version of the paragraph where John took off his sweatshirt than in a version where he put it on. They argued that such a result indicates that, when John is said to put on his sweatshirt, the reader constructs a mental model with John wearing the sweatshirt. Since the paragraph is about John, John and his sweatshirt will both be part of the mental model that the reader constructs, even though the sweatshirt is never mentioned again. But if John had taken the sweatshirt off, the reader no longer includes it in the updated mental model, making it less available to memory.

Bower and Morrow (1990) presented probes during the reading of a passage to show that situation models containing spatial relations were constructed even while a passage was being read. Their participants first memorized a map describing the linear pathways through a series of rooms, together with the objects that were in the rooms. They then read a text that described the movement of a character through the rooms. Their reading was interrupted by the presentation of a pair of object names, and they were to decide whether or not the two objects were in the same room. Decisions were slower when the objects were further away from the room where the character was when the text was interrupted, strongly suggesting that participants were mentally representing the movement of the character through the mental map of the rooms.

Situation models can also represent the perspective that protagonists have on the situation. O’Brien and Albrecht (1992) had participants read passages like 9.37 (some with, some without, several sentences intervening between the initial and the target sentence):

As Kim stood inside/outside the health club, she felt a rush of excitement. (. . .)

She was getting anxious to start and was glad when she saw the instructor come
in the door of the club. . . . (9.37)

Line-by-line reading time was measured. The critical line included the information about the instructor coming in the door of the club. This is consistent with the perspective of the protagonist, Kim, when the first sentence said she was inside the club; it is inconsistent when the first sentence put Kim outside the club. Participants read the consistent line faster than the inconsistent line—but only when they were told that they should imagine that they were doing whatever the main character was doing. Thus, participants could take the perspective of that character, but they did not automatically do so.

This finding brings up the fact that constructing a situation model can take time and processing capacity. We have actually seen examples of this earlier, in our discussion of Mandler's (1986) and Zwaan's (1996) findings that passages are comprehended more quickly when events are described in the order in which they occur and when events that are described together take place close in time to one another. Time seems to be represented in situation models, and representing time is easier when the text maps naturally on to time relations.

An interesting experiment that appears to show influences of both lexical relations and of mental models was reported by Garrod and Terras (2000) (note, these authors did not discuss their data in terms of mental models). They measured eye movements while participants read passages that contained short texts like 9.38, after introducing a referent for the initial pronoun (e.g., a teacher):

She was busy writing a letter of complaint to a parent. However, she was
disturbed by a loud scream from the back of the class and the pen dropped
on the floor. (9.38a)

She was busy writing a letter of complaint to a parent. However, she was
disturbed by a loud scream from the back of the class and the chalk dropped
on the floor. (9.38b)

She was busy writing an exercise on the blackboard by the door. However, she
was disturbed by a loud scream from the back of the class and the pen dropped
on the floor. (9.38c)

She was busy writing an exercise on the blackboard by the door. However, she
was disturbed by a loud scream from the back of the class and the chalk dropped
on the floor. (9.38b)

There were also versions of the first sentence that explicitly mentioned a pen or chalk.

Facilitating effects of the lexical relation between *writing* and the typical instrument *pen* were apparent in measures of early reading on *pen dropped* and *chalk dropped* in the second sentence: *pen dropped* was read equally quickly whether or not a pen had been explicitly mentioned in the preceding text, but *chalk dropped* was read faster when chalk had been previously mentioned. Garrod and Terras suggested that this effect reflects a process of “bonding” an anaphor with a lexically related antecedent. Of more relevance to the current discussion of mental models, however, is the following. When the instrument was not explicitly mentioned in the first sentence, go-past time on the verb *dropped* was substantially longer for *pen dropped* following the blackboard context than following the letter of complaint context, and second-pass time on *pen* was longer as well.

While the go-past difference did not appear following *chalk dropped*, second-pass times on *chalk* were longer when *chalk dropped* had appeared in the letter of complaint context than in the blackboard context. Garrod and Terras referred to a process of resolution underlying these later effects. It appears that readers who read about writing a letter or writing on a blackboard form some sort of mental model with a plausible instrument, and text that is congruent with the typically formed mental model is read more fluently.

Readability

One would hope that, if we fully understood the process of reading and understanding texts, we would know how to construct texts that are easily understood—texts that are highly readable. Perhaps we could list all the aspects of text that can plausibly affect reading difficulty, starting with lexical variables such as the frequency of the words in the language, the length of the words, moving through syntactic variables measuring complexity of sentences such as the length of sentences, number of phrases in a sentence, number of clauses in a sentence, and then moving to more discourse-level variables such as the number of propositions, the complexity of the causal structure, etc. Then we could take all of these measures and see how well the combination can predict how difficult readers find the text.

Presumably, if you can come up with objective measures for each of the above predictor variables, and some combination of those variables in fact do a pretty good job of predicting how difficult readers find the text, then it would appear there is some sense in which you can say that you understand discourse processing. In addition, if you can find out which variables are doing most of the prediction, you can understand which text variables are really important in understanding discourse. Moreover, one would have a method for predicting difficulty of text that should have implications for helping to design educational curricula and other practical applications.

There is in fact, a reasonably large and fairly old literature on the topic (for reviews see Chall, 1958; Kintsch & Vipond, 1979). Most of the impetus was practical; people were trying to use objective techniques for measuring the relative difficulty of different texts for educational purposes. More recently, as correlational methods have become more sophisticated, the emphasis has been more on trying to understand which variables are the most important. While this enterprise is of some value, we think that the conclusions that can be drawn are limited. In this correlational procedure a set of passages are taken from various texts (let's say there are 50 such passages), and various objective indices, such as average word length, average word frequency, average sentence length, average number of phrases per sentence, etc., are computed for each of the paragraphs. Then a group of participants are asked to read the passages and some measure of reading difficulty (usually the average reading speed in words per minute) is computed for each passage. Finally, the reading times for the 50 passages are correlated with each of the objective indices one has computed, and the pattern of correlations is analyzed by a technique known as multiple regression, in order to achieve enlightenment.

One problem with the technique may already be apparent: why use reading speed as a measure of how much difficulty the reader is having? Presumably a measure of the reader's comprehension of the passage is needed as well. This leads to a very thorny issue: How does one measure comprehension? As you are probably aware from taking reading comprehension tests of various kinds, most reading comprehension questions tap many things besides your comprehension of the text. Testing readers for recognition of detail, such as dates and names of minor characters, does not seem to be testing for what one would want to call comprehension. Further, some people can answer questions that ask about content on the basis of prior knowledge, without much contribution from the passage.

In other words, if one wants to measure comprehension, it cannot be done in a vacuum; one needs a theory of what comprehension is (Kintsch & Vipond, 1979). Perhaps the solution is to use several measures of comprehension, each tapping a different facet of comprehension. For example, one would measure how many essential points of the story or passage are remembered, how much of the causal structure is remembered, the number of propositions that are recalled, how much lexical detail is remembered, etc. The problem with doing that is that most of the reason for the enterprise is then lost, since the goal we started with was to determine which variables are important for some global measure of readability. As a result, many experiments (e.g., Kintsch & Vipond, 1979) have measured readability using both reading speed and comprehension in terms of a memory test, in which memory has typically been measured in terms of something like number of idea units or propositions recalled (inspired by theories such as Kintsch & van Dijk's).

Even if one had the perfect behavioral measure of reading difficulty, there are still major difficulties in drawing any firm conclusions from the method. The basic one is that the method is correlational. As you undoubtedly know, it is difficult to draw causal inferences from correlational data. We will illustrate the problem by reviewing some of the findings. Many different educational researchers have come up with readability formulas, 50 or more of them (Kintsch & Vipond, 1979). Most are general-purpose, but some are intended to apply only to specialized areas such as chemistry texts. A major aspect of all of these measures is that simple low-level variables (especially the average word length and the number of words per sentence) are important components of the formulas, since they are highly predictive of how difficult the text is. (Usually, the average length of a word is the most predictive.) What can one infer from that? Unfortunately, very little.

First, let us take the average length of a word. As we indicated in Chapter 3, the average length of a word is highly correlated (negatively) with its frequency in the language. Thus, one cannot be sure whether the relation is due to the fact that longer words have more visual information, or because longer words are less frequent and thus harder to locate in the lexicon (see Chapter 3). There is no way to understand from correlational data which aspect is truly operative, or whether both are. (In contrast, as discussed in Chapter 1, one can experimentally control one variable and vary the other to get some idea whether the variables operating separately each have some effect.) The problem is worse when considered in the context of prose passages, however, since texts with longer words will also tend to have longer and more complex sentences, be expressing more complex ideas, etc. While multiple correlation techniques allow you to understand something about whether a variable has any effect when the effect of another is taken into account, it does not really allow you to say very much.

One reason why the low-level variables tend to do much of the predicting is that they are probably more reliable indices of the underlying psychological variables that they are measuring than are the higher-order ones. Word length is a reliable measure of the amount of visual information needed to be extracted in the word and also, since it is highly correlated with word frequency, a reliable predictor of lexical access time. In contrast, a complexity measure derived from existing analyses of discourse structure is probably at best capturing only part of what one would want to mean as "discourse complexity."

Another way to view the problem with the correlational method is to think about readability measures as a guide to the writer. Let's say that we know that text with shorter words and shorter sentences tends to be more readable than text with longer words and longer sentences. Does that mean that if we have a text with long words and long sentences, we should go about rewriting it so as to make both word and sentence length shorter? Perhaps, to some extent. If we have statements like *eschew obfuscation* and run-on sentences, it might be a good idea to change them. However, the underlying ideas that one is attempting to express in difficult text are often inherently complex.

Thus longer sentences and more-complex sentence structures are probably needed to explain these ideas, and they will take longer to read than simpler sentences expressing simpler ideas. Using simpler sentences will probably either make the text incomprehensible (if one is still trying to express the same ideas) or simplify the text at the expense of removing many of the ideas one was originally trying to express.

To summarize, readability formulas are of some value in allowing an educator to predict how difficult a group of participants will find a given text. However, the correlational data from which they are obtained are not of much help in understanding discourse processing. It is possible the situation may change as our understanding of the reading process increases and gets incorporated into theoretically motivated indices of readability. McNamara et al. (2010) present one example of a step in this direction. They describe a computational technique for integrating various measures of text cohesion and coherence with lower-level variables and using the result to predict comprehension of text. Another recent development with some promise has been to predict moment-to-moment processing rather than global text comprehension. For example, the gaze duration on a word might be the variable that one is trying to predict, and then one would use indices such as word length, sentence length, position in sentence, etc. to predict gaze time. Using smaller units allows one to be more diagnostic, but does require accepting the “immediacy hypothesis” (i.e., that the time spent on a word reflects all the processing done on that word and only that word). As we discussed in Chapters 4 and 8, strong versions of the immediacy hypothesis overstate the actual facts and may seriously misrepresent the processing of complex discourse variables.

Summary

In previous chapters we discussed how readers recognize words, access their meaning, and use their knowledge of the structure of their language to put these meanings together in the process of understanding sentences. It is possible to provide fairly clear and explicit descriptions of the knowledge that readers bring to these tasks and to develop fairly clear and explicit theories of how they use this knowledge.

The present chapter discussed how readers put sentence meanings together into coherent representations of texts. The knowledge they use to do this is far more extensive, more amorphous in fact, than their knowledge of the lexicon and the grammar of their language, and theories of how they use this knowledge are accordingly far less constrained than theories of word recognition and sentence comprehension. Nonetheless, discourse comprehension researchers have been able to make substantial progress in understanding the process of comprehending texts. They have learned a good deal about how readers use anaphoric references to link sentences together. They have uncovered interesting facts about how readers use connectives like *because* and *later* to relate propositions in the text to one another and how they use the information structure of sentences (e.g., their given–new structure) to this end. Researchers have studied how memory for earlier material is accessed to make sense of what is currently being read. They have studied how readers go beyond what is literally said and make inferences that are required to tie a discourse together, and how they can (and sometimes do) make such inferences to enrich the description that the text provides. And they have shown how readers go beyond building networks of propositions and construct mental models that capture what anyone would want to call the meaning of a text.

We have to end with a cautionary note, however. The processes involved in comprehending text and making mental models of what is read use, in principle, all of our knowledge. And we have suggested that a mental model of a text can represent whatever a person knows about a situation. It is a daunting task to even contemplate building explicit models of what we know and how we use it (perhaps not even a possible task; J. A. Fodor, 1983; for discussion of problems and possible

solutions see Koppen, Noordman, & Vonk, 2008). We have to accept that our theories of mental models and inference process, etc., are incomplete at best, and often vague and inexplicit. But even so we can look back at the research that has been done on text comprehension and know that it has taught us much that we did not previously know, and we can look ahead to future research with the same expectation in mind.